MAS508-G Control Theory
Lecture 2: System Modelling

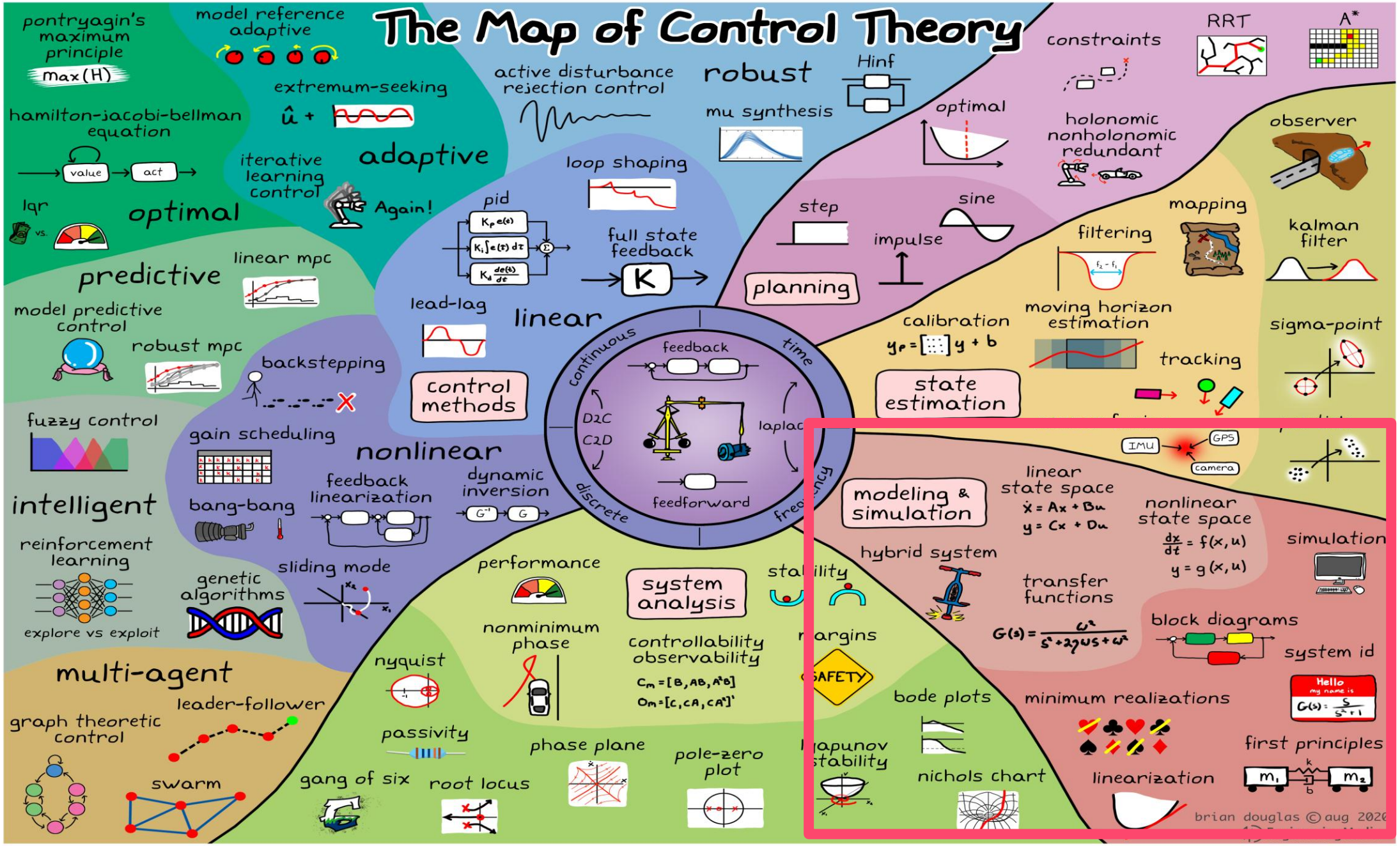
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Website: https://www.uia.no/om-uia/ansatte/jingz


The Map of Control Theory
pontryagin's maximum principle
Max(H)
model reference adaptive
extremum-seeking
hamilton-jacobi-bellman equation
value
act
iterative learning control
Again!
adaptive
optimal
lqr
vs.
active disturbance rejection control
robust
Hinf
mu synthesis
loop shaping
pid
full state feedback
K
linear
lead-lag
control methods
predictive
linear mpc
model predictive control
robust mpc
backstepping
fuzzy control
gain scheduling
nonlinear
feedback linearization
dynamic inversion
bang-bang
intelligent
reinforcement learning
explore vs exploit
genetic algorithms
sliding mode
multi-agent
graph theoretic control
leader-follower
swarm
continuous
discrete
time
freq
D2C
C2D
feedback
feedforward
laplac
constraints
RRT
A*
optimal
holonomic
nonholonomic
redundant
observer
step
sine
impulse
planning
filtering
mapping
kalman filter
calibration
moving horizon estimation
tracking
sigma-point
state estimation
IMU
GPS
camera
modeling & simulation
linear state space
nonlinear state space
simulation
hybrid system
transfer functions
block diagrams
system id
minimum realizations
first principles
linearization
bode plots
nichols chart
lyapunov stability
SAFETY
performance
system analysis
nonminimum phase
controllability
observability
nyquist
passivity
phase plane
pole-zero plot
gang of six
root locus
brian douglas © aug 2020

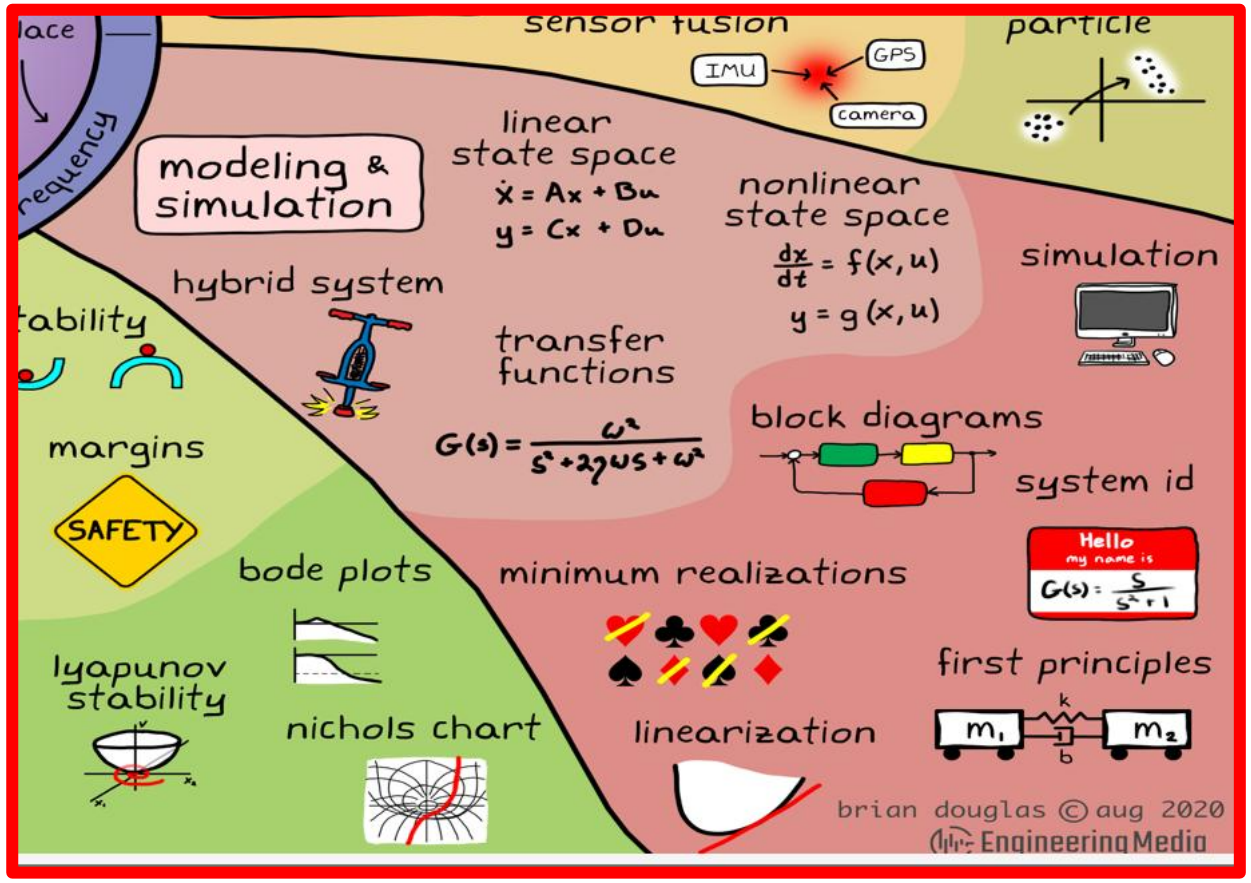
Lecture 2 System Modeling

a) **Review of Theoretical Modeling**
   - Differential equations
   - Transfer functions
   - Block diagrams
   - Frequency response

b) **State-Space Representation**

c) **Experimental Modeling**

d) **Linearization**



Types of Representations

$$\frac{J_M d\omega(t)}{dt} + D_m\omega(t) = T_M(t)$$

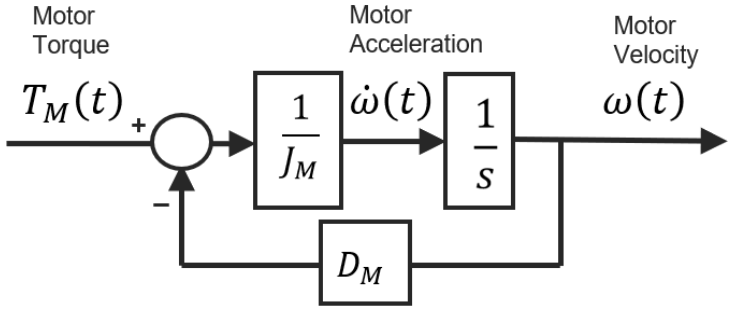
Differential equation

$$\frac{\Omega(s)}{T_M(s)} = \frac{1}{J_M s + D_M}$$

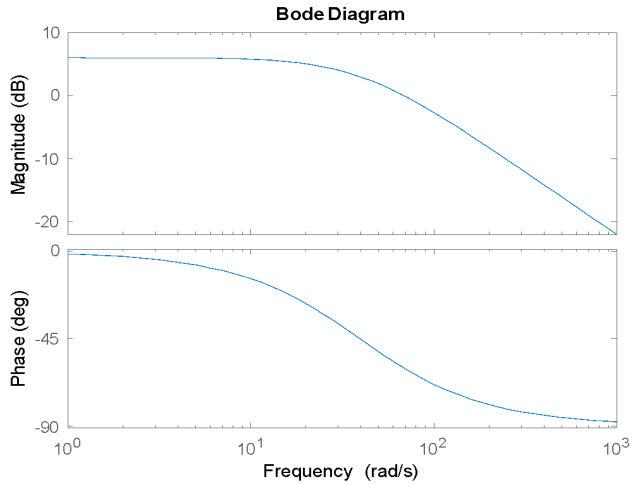
Transfer function

$$\dot{x} = Ax + Bu$$
$$y = Cx + Du$$

State-space



Block diagram



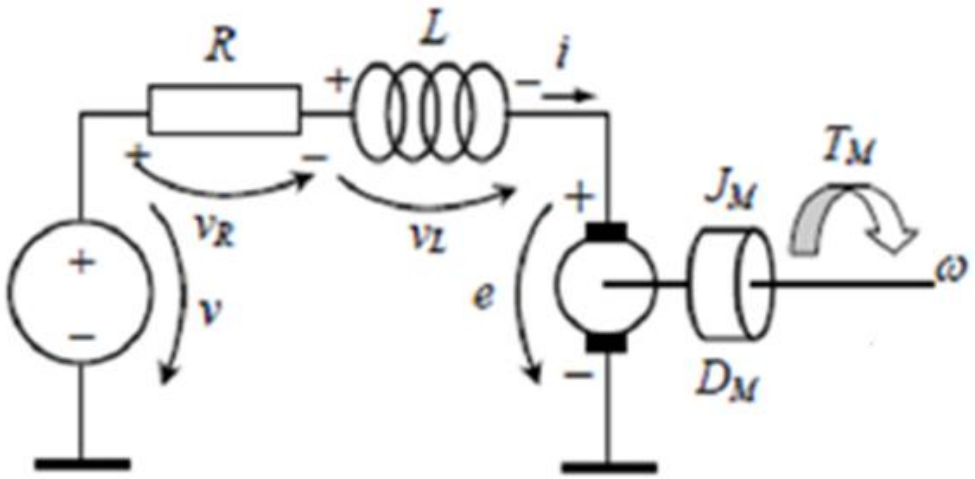
Frequency response

Lecture 2a Theoretical Modelling

- Differential equations
- Transfer functions
- Block diagrams
- Frequency response

What is a model?

Mathematical system that represents the dynamics of a system



DC motor



$$\frac{J_M d\omega(t)}{dt} + D_m\omega(t) = T_M(t)$$

Differential equation

$$\frac{\Omega(s)}{T_M(s)} = \frac{1}{J_M s + D_M}$$

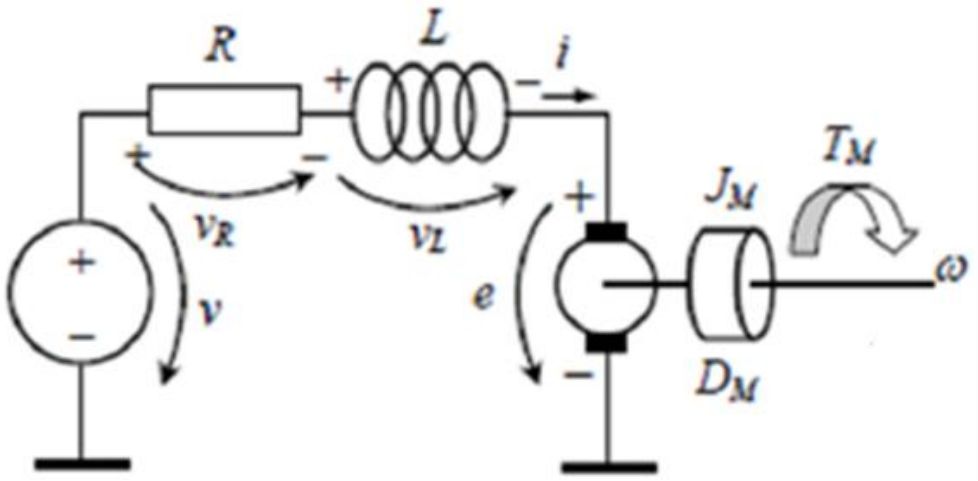
Transfer function

Differential equations

Book “Control Systems Engineering”, Chapter 2

What is theoretical modelling?

- Modelling a system using equations based on physical principles
- Obtains a mathematical representation of system dynamics



DC motor



$$\frac{J_M d\omega(t)}{dt} + D_M \omega(t) = T_M(t)$$

Differential equation

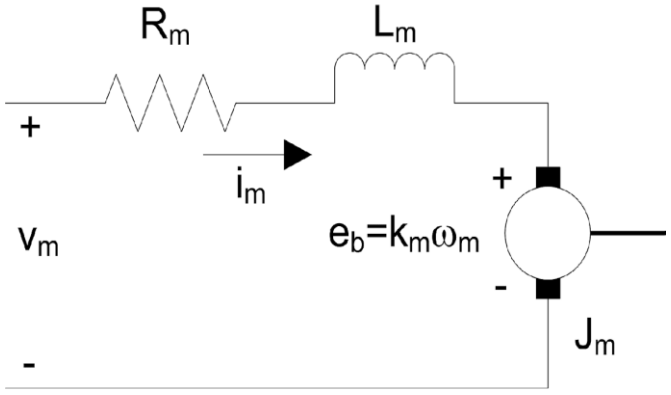
Model Derivation Methods

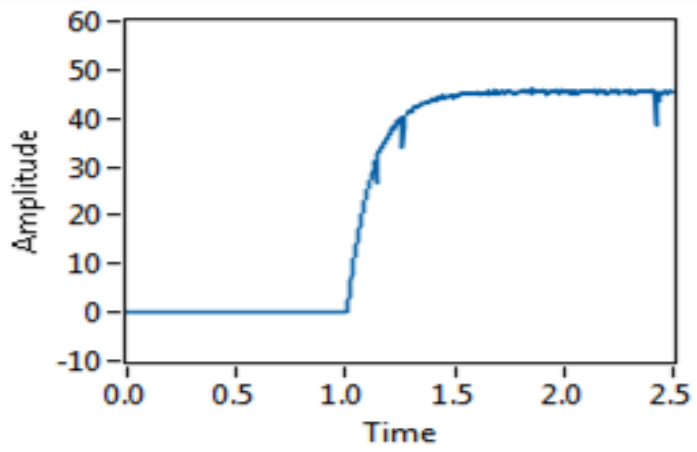
Analytical

- First-principles, such as using Newton's law, KVL, KCL, Mass-balance, Lagrange
- Block diagram

Experimental

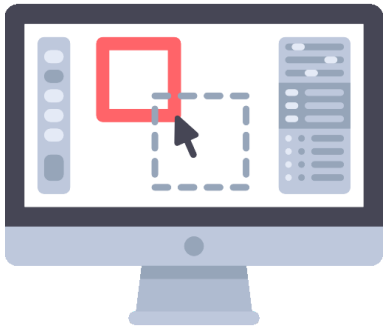
- Step response / Bump test
- Frequency Response





Theoretical Modelling

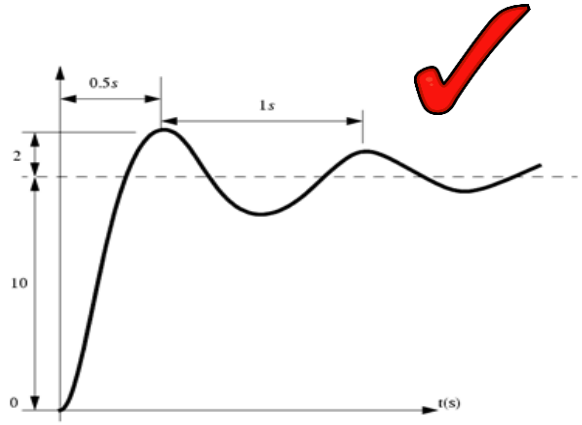
- Theoretical modelling stresses the importance of "**knowing your system before you control it**"
- With a model, You can design a controller
  - when hardware not available yet and verify controller in simulation before implementation on hardware
  - that meets the requirements faster and with better performance
- Sometimes it's necessary
  - Not all hardware can be modeled experimentally, e.g. unstable or potentially dangerous systems



Design

Plant Model

$$G_s(p) = \frac{b_m p^{\beta_m} + \ldots + b_1 p^{\beta_1} + b_0 p^{\beta_0}}{a_n p^{\alpha_n} + \ldots + a_1 p^{\alpha_1} + a_0 p^{\alpha_0}}.$$



Electro-mechanical: DC Motor

The DC motor is a hybrid of electrical and mechanical variables. The electric equivalent circuit of the armature and the free-body diagram of the rotor are shown in the following figure. The input of the system is the voltage source $v(t)$ applied to the motor's armature, while the output is the rotational speed of the shaft $\omega(t)$.



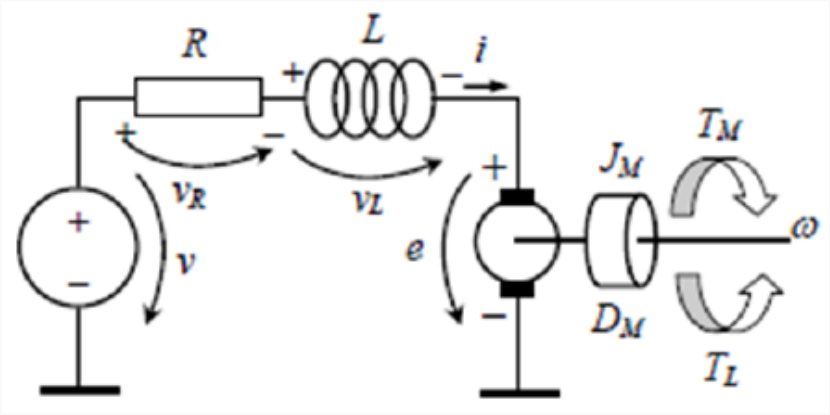


Figure: Motor

- $v(t)$ $[V]$: applied voltage
- R $[\Omega]$: armature resistance
- L $[H]$: armature inductance
- $i(t)$ $[A]$: armature current
- $e(t)$ $[V]$: induced voltage
- $T_M(t)$ $[Nm]$: motor torque
- $T_L(t)$ $[Nm]$: load torque
- J_M $[kgm^2]$:moment of inertia of the engine/motor
- D_M $[Nm/(rad/s)]$: viscous damping constant
- K_e $[V/(rad/s)]$: tension constant
- K_T $[Nm/A]$: torque constant
- $\omega(t)$ $[rad/s]$: rotational speed

DC motor model - System equations

Kirchhoff Voltage Law(KLV) for the circuit:

$$v(t) - Ri(t) - L\frac{di(t)}{dt} - e(t) = 0 \qquad \text{(s1)}$$

The induced voltage is proportional to the rotational speed:

$$e(t) = K_e\omega(t) \qquad \text{(s2)}$$

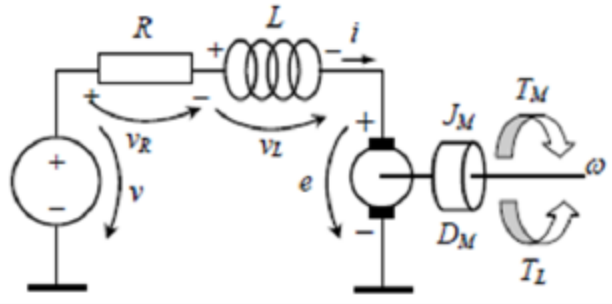


Figure: Motor

Moment balance for the rotational-mechanical system:

$$J_M\frac{d\omega(t)}{dt} = T_M(t) - D_M\omega(t) - T_L(t)$$

Assume Electrical motor without load, such as $T_L = 0$

$$J_M\frac{d\omega(t)}{dt} = T_M(t) - D_M\omega(t) \qquad \text{(s3)}$$

The torque developed by the motor is proportional to the armature current.

$$T_M(t) = K_T i(t) \qquad \text{(s4)}$$

DC Motor equations

$$v(t) - R\,i(t) - \frac{Ldi(t)}{dt} - e(t) = 0 \qquad (s1)$$

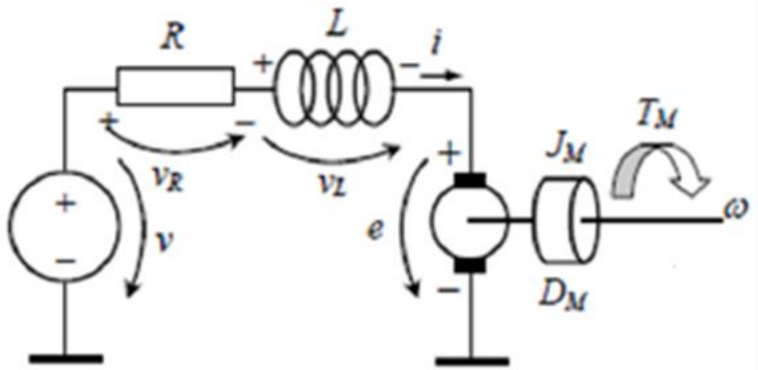
$$e(t) = K_e\omega(t) \qquad (s2)$$

$$\frac{J_M d\omega(t)}{dt} + D_M\omega(t) = T_M(t) \qquad (s3)$$

$$T_{M(t)} = K_T i(t) \qquad (s4)$$

DC motor





Theoretical Modelling: Takeaways

Benefits

- Gain more in-depth understanding of system dynamics
- Necessary for certain system (e.g. unstable, dangerous ones)
- Can model a system with hardware not available or not accessible, e.g. to simulate, design a controller

Drawbacks

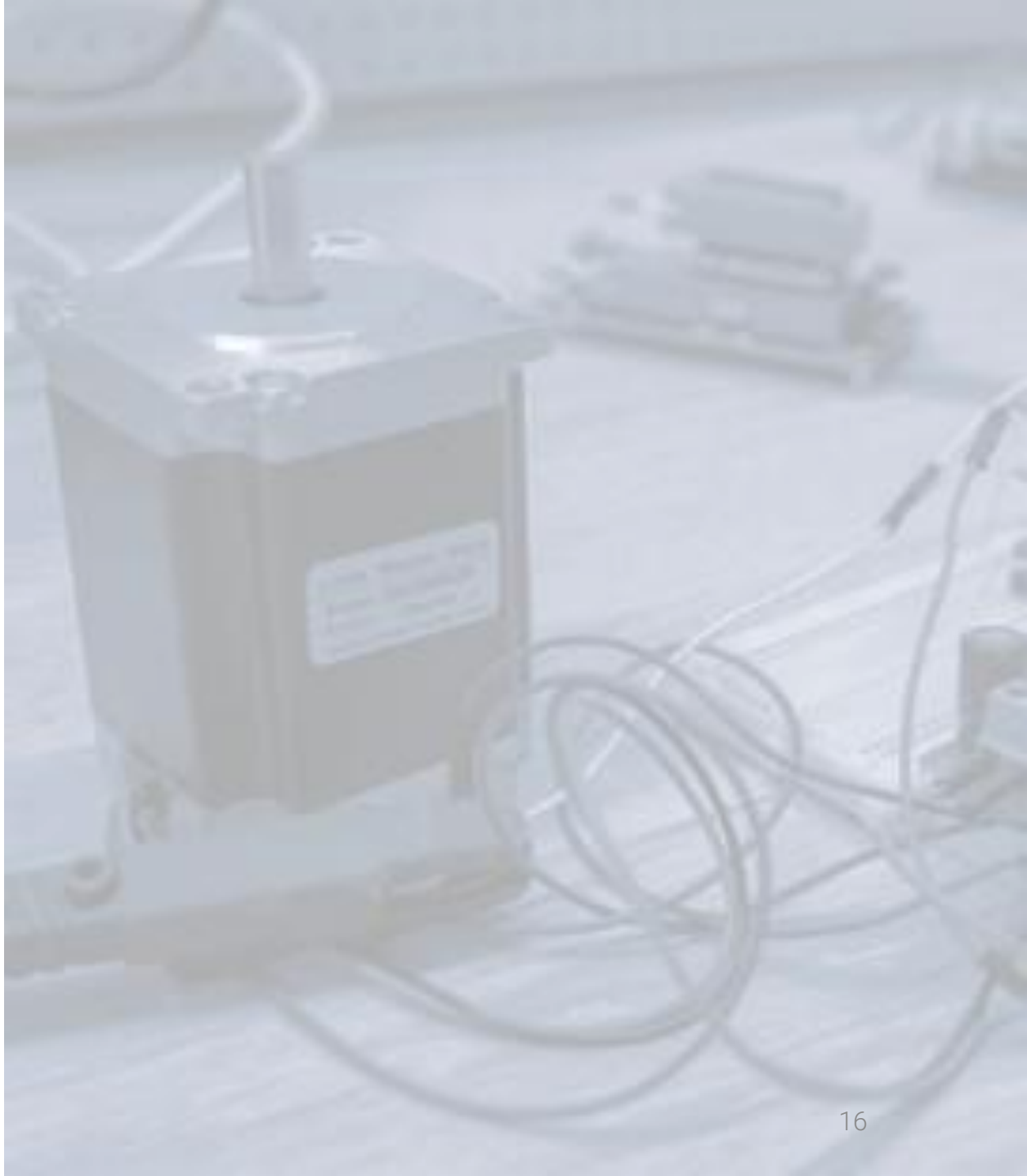
- Needed accurate system specifications
- May not take into account all the dynamics of a system (e.g. viscous friction)
- Can be tedious process for some systems

Transfer Functions

Book “Control Systems Engineering”, Chapter 2

Transfer Function

- Represent the input–output relationship for a Linear Time Invariant System
- Laplace transform yields transfer functions that allow for different types of system analysis
- In control systems, transfer functions are OFTEN used for modelling systems
- … and designing controllers, e.g. to stabilize systems.



Transfer function

- Laplace Transform converts **a linear time-invariant differential equation** into the frequency or "S-domain"
- Laplace Transform with **Zero initial conditions**

$$\frac{dc(t)}{dt} + 2c(t) = r(t)$$

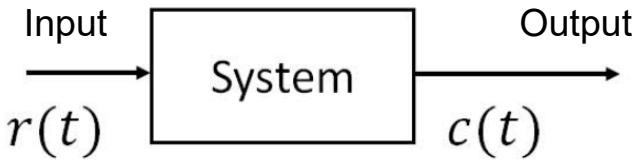
Laplace transform (Zero initials)

$$sC(s) + 2C(s) = R(s)$$

Transfer function

$$G(s) = \frac{C(s)}{R(s)} = \frac{1}{s+2}$$

$$\mathrm{G(s)} = \frac{\mathrm{Output}(s)}{\mathrm{Input}(s)} = \frac{Y(s)}{R(s)}$$





Transfer Function of Linear Differential Equation

Linear differential equation with input u(t) and output y(t)

$$a_n \frac{d^n y(t)}{dt^n} + a_{n-1} \frac{d^{n-1} y(t)}{dt^{n-1}} + \cdots + a_0 y(t) = b_m \frac{d^m u(t)}{dt^m} + b \frac{d^{m-1} u(t)}{dt^{m-1}} + \cdots + b_0 u(t)$$

Assuming initial conditions are zero, this can be simplified to:

$$a_n s^n Y(s) + a_{n-1} s^{n-1} Y(s) + \cdots + a_0 Y(s) = b_m s^m U(s) + b_{m-1} s^{m-1} U(s) + \cdots + b_0 U(s)$$

Transfer function G(s):

$$G(s) = \frac{Y(s)}{U(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_0}$$

DC Motor Transfer Function

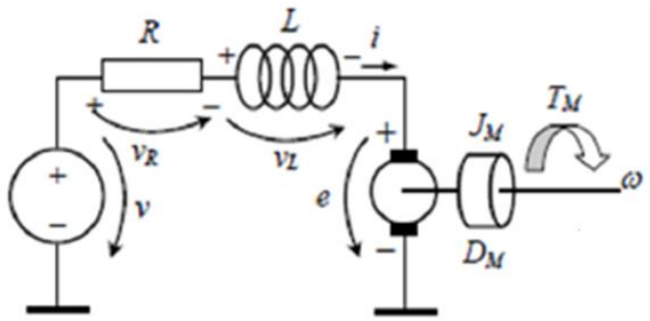
$$v(t) - R\,i(t) - \frac{Ldi(t)}{dt} - e(t) = 0 \qquad (s1)$$

$$e(t) = K_e\omega(t) \qquad (s2)$$

$$\frac{J_M d\omega(t)}{dt} + D_M\omega(t) = T_M(t) \qquad (s3)$$

$$T_{M(t)} = K_T i(t) \qquad (s4)$$





- Input: Applied voltage $v(t)$
- Output: Motor rotational speed $\omega(t)$

- Transfer function: $G(s) = \frac{\Omega(s)}{V(s)}$

DC Motor Transfer Function

Insert $e(t) = K_e\omega(t)$ and $T_M(t) = K_T i(t)$ to Moment balance and KVL circuit equations:

$$\begin{aligned} J_M \frac{d\omega(t)}{dt} + D_M\omega(t) &= K_T i(t) \\ Ri(t) + L\frac{di(t)}{dt} + K_e\omega(t) &= v(t) \end{aligned}$$

Take Laplace transform:

$$\begin{aligned} (J_M s + D_M)\Omega(s) &= K_T I(s) \\ (Ls + R)I(s) &= V(s) - K_e\Omega(s) \end{aligned}$$

Calculate $I(s) = \frac{(J_M s + D_M)\Omega(s)}{K_T}$ from the first equation and insert it to the second equation:

$$(Ls + R)(J_M s + D_M)\Omega(s) + K_T K_e\Omega(s) = K_T V(s)$$

Transfer function

$$G(s) = \frac{\Omega(s)}{V(s)} = \frac{K_T}{(Ls + R)(J_M s + D_M) + K_e K_T} = \frac{K_T/(LJ_M)}{s^2 + \frac{(RJ_M + LD_M)}{LJ_M}s + \frac{(RD_M + K_e K_T)}{LJ_M}}$$

Transfer Function: Takeaways

Benefits

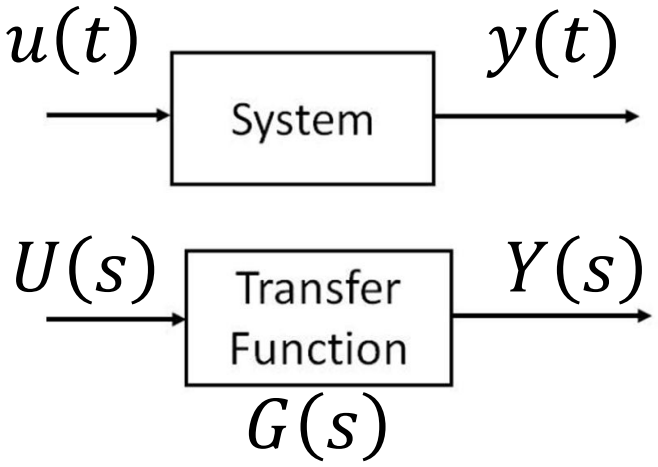
- Transfer function-based models are essential to classic control design
- Simple representation of a process dynamics.
- Makes a lot of analysis easier

(Easier to manipulate and analysis than ODE)

Drawbacks

- Cannot be used for nonlinear systems
- Not ideal for modelling MIMO systems if you are planning on designing a controller.

$$\mathrm{G(s)} = \frac{\mathrm{Output}(s)}{\mathrm{Input}(s)} = \frac{Y(s)}{U(s)}$$

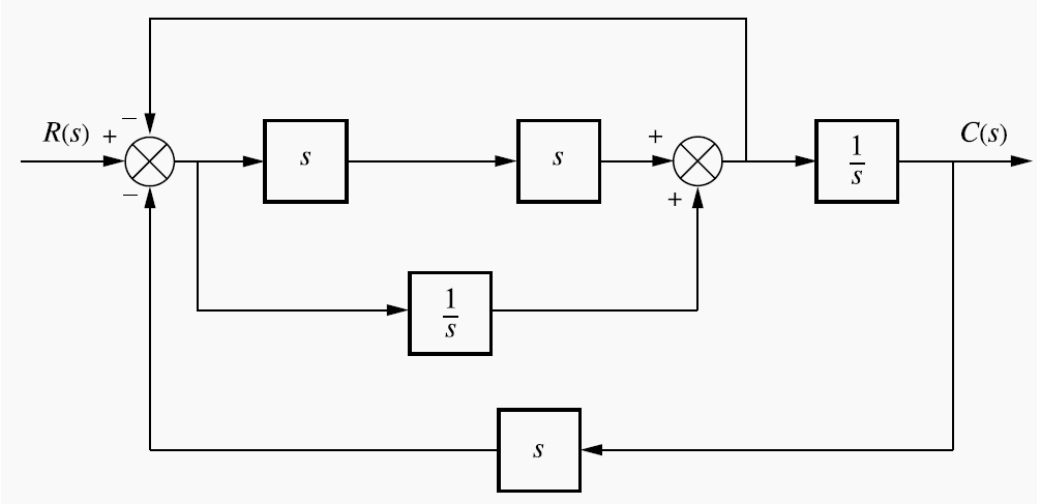


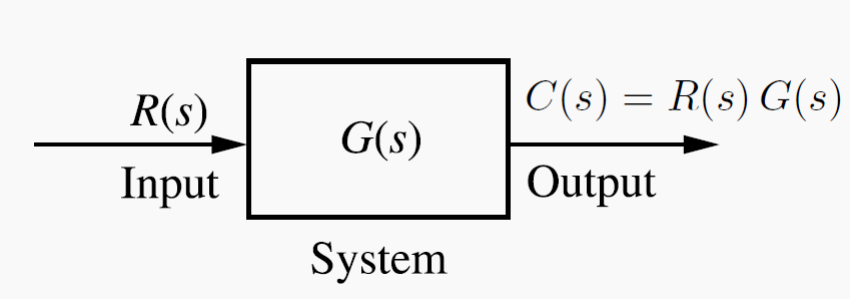
Block Diagrams

Book “Control Systems Engineering”, Chapter 5

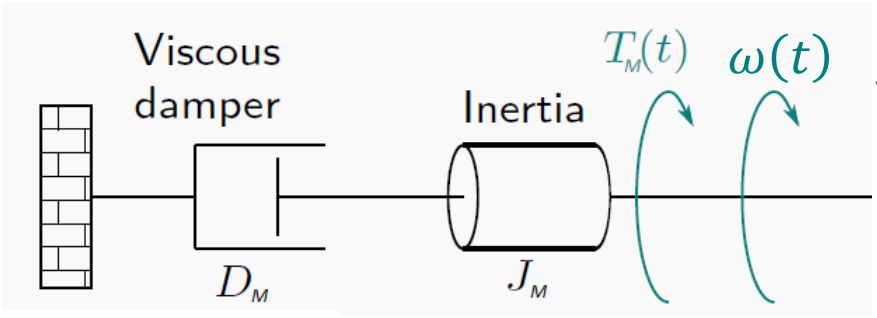
What is a block diagram?

- Show the interaction between different system components
- Block diagram provides a pictorial representation of a system
- Operational block representing individual system
- Three basic elements:
  - Rectangles → Operators
  - Lines → Signals
  - Circles → Addition and Subtraction





Block Diagram



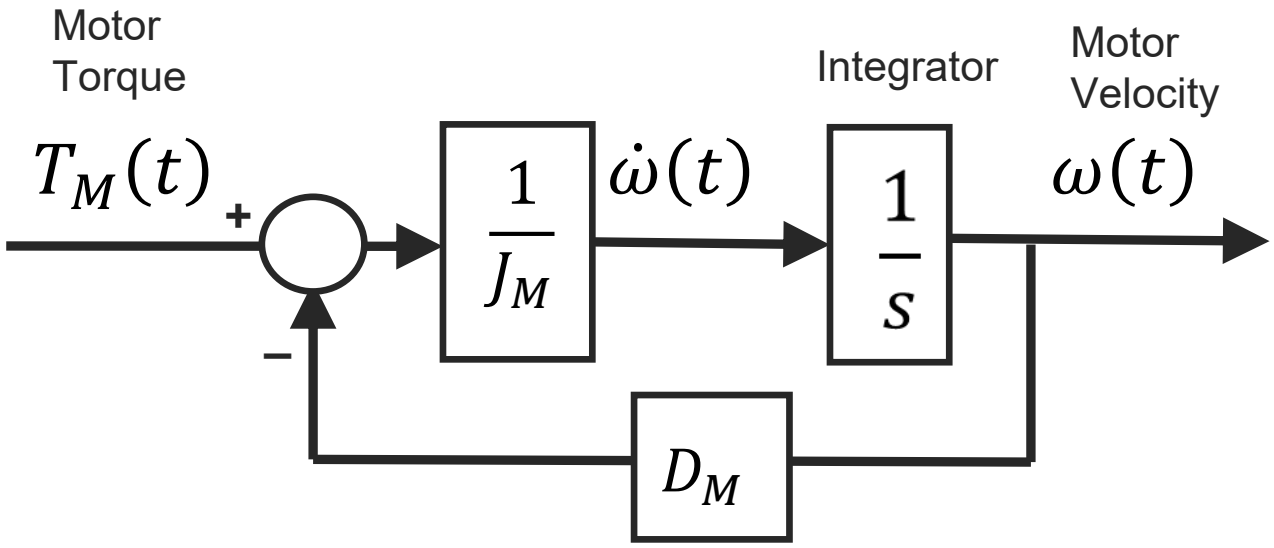
- *Motor – rotational mechanical*

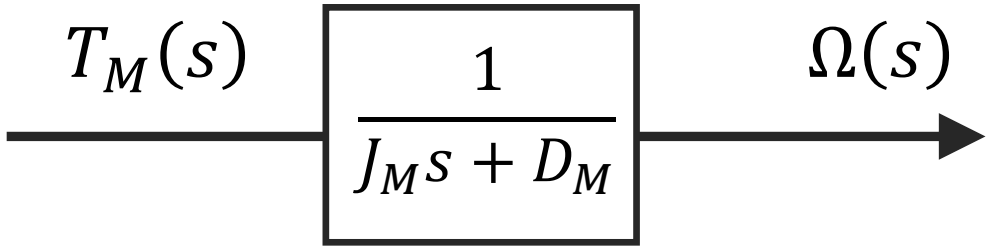
Equation: $\frac{J_M d\omega(t)}{dt} + D_M\omega(t) = T_M(t)$

Transfer function:

$$\frac{\Omega(s)}{T_M(s)} = \frac{1}{J_M s + D_M}$$

Block diagram:





Why do you need a block diagram?

- To model a system
- Visually show how different components interact with each other
- To design a control system
- Process elements are shows as boxes with inputs and outputs
- Inputs are the variables that influence the process
- Outputs are the resulting response or outcome of the process



DC Motor equations

$$v(t) - R\,i(t) - \frac{Ldi(t)}{dt} - e(t) = 0 \qquad (s1)$$

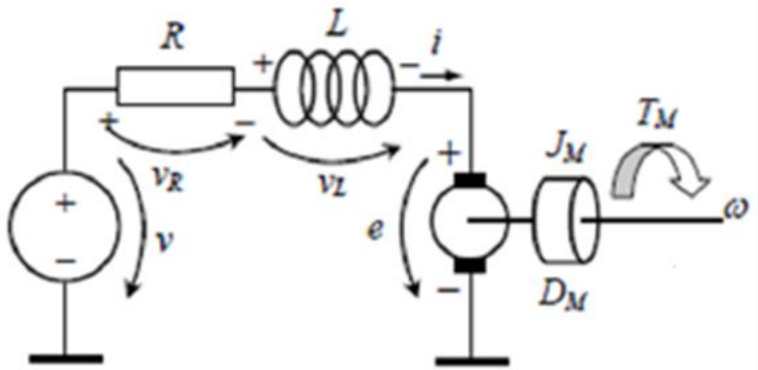
$$e(t) = K_e\omega(t) \qquad (s2)$$

$$\frac{J_M d\omega(t)}{dt} + D_m\omega(t) = T_M(t) \qquad (s3)$$

$$T_{M(t)} = K_T i(t) \qquad (s4)$$

DC motor





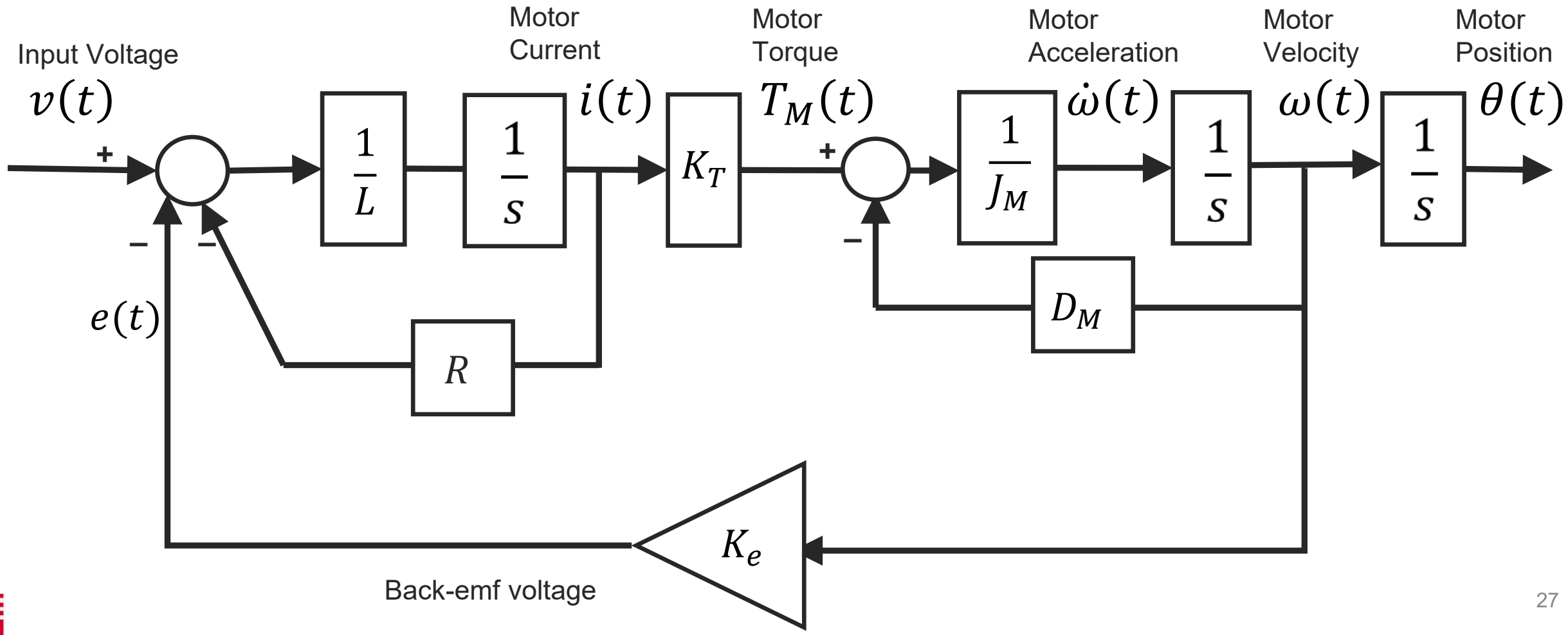
DC Motor Block Diagram Example - based on equations

$$v(t) - R\,i(t) - \frac{L di(t)}{dt} - e(t) = 0 \qquad (s1)$$

$$e(t) = K_e \omega(t) \qquad (s2)$$

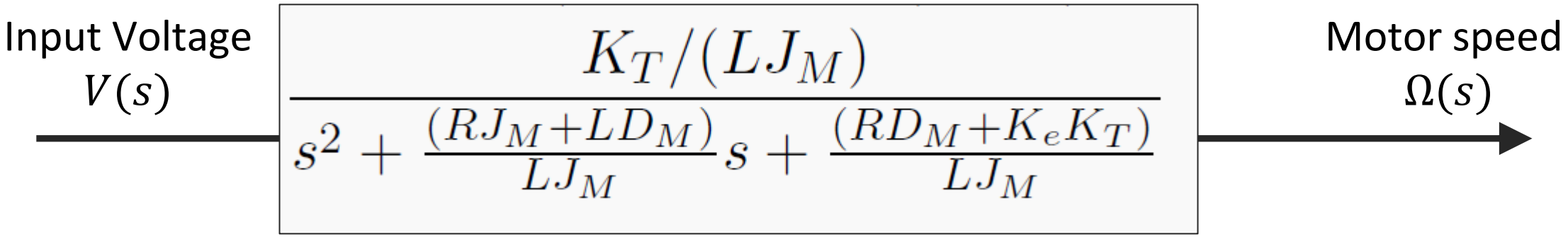
$$\frac{J_M d\omega(t)}{dt} + D_m \omega(t) = T_M(t) \qquad (s3)$$

$$T_{M(t)} = K_T i(t) \qquad (s4)$$



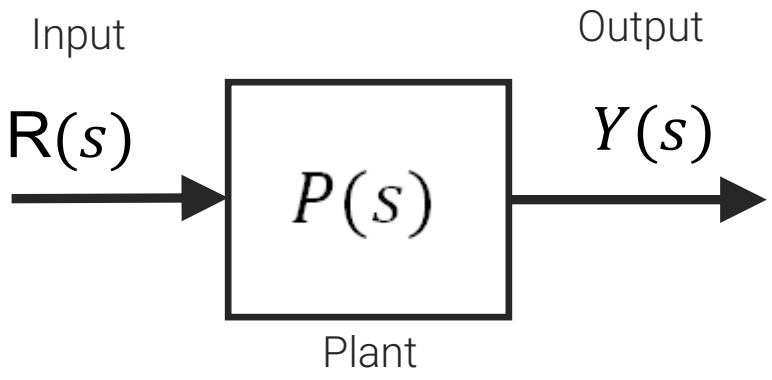
DC Motor Block Diagram Example

- based on transfer function

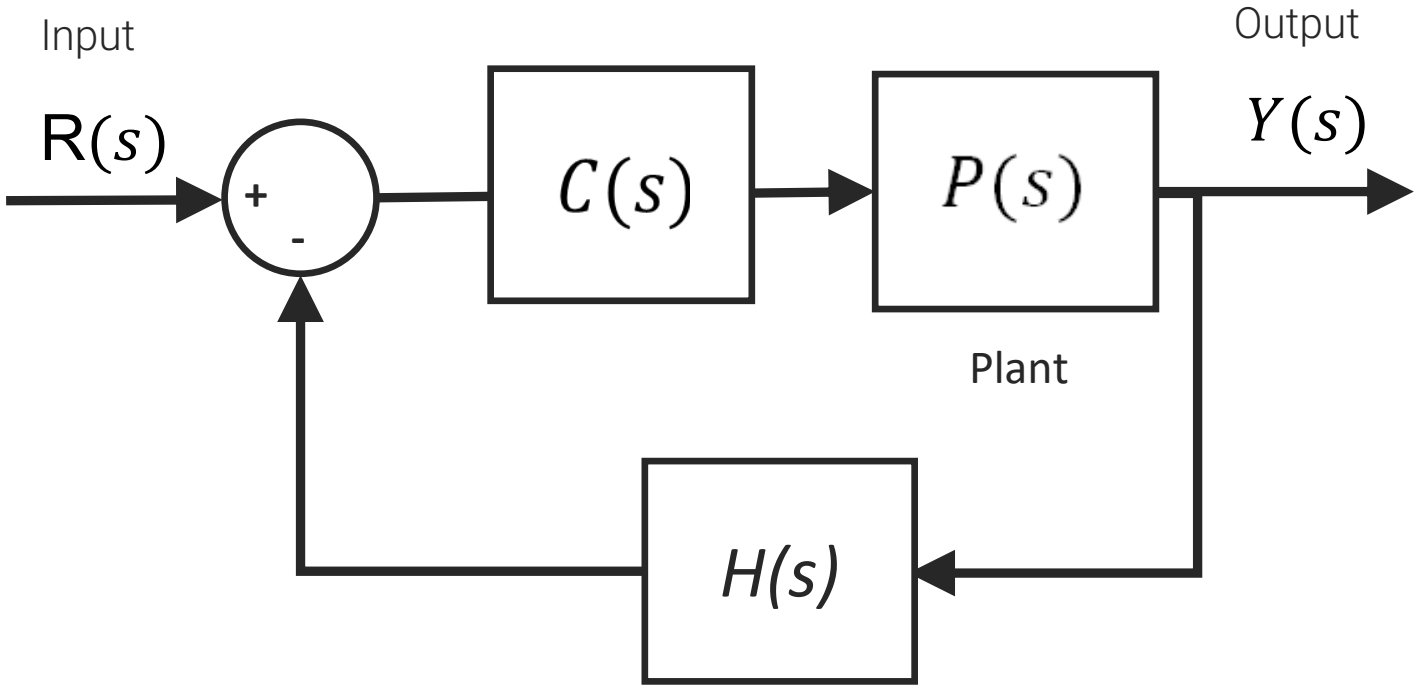


Open-Loop and Closed-Loop Systems

Open-loop system

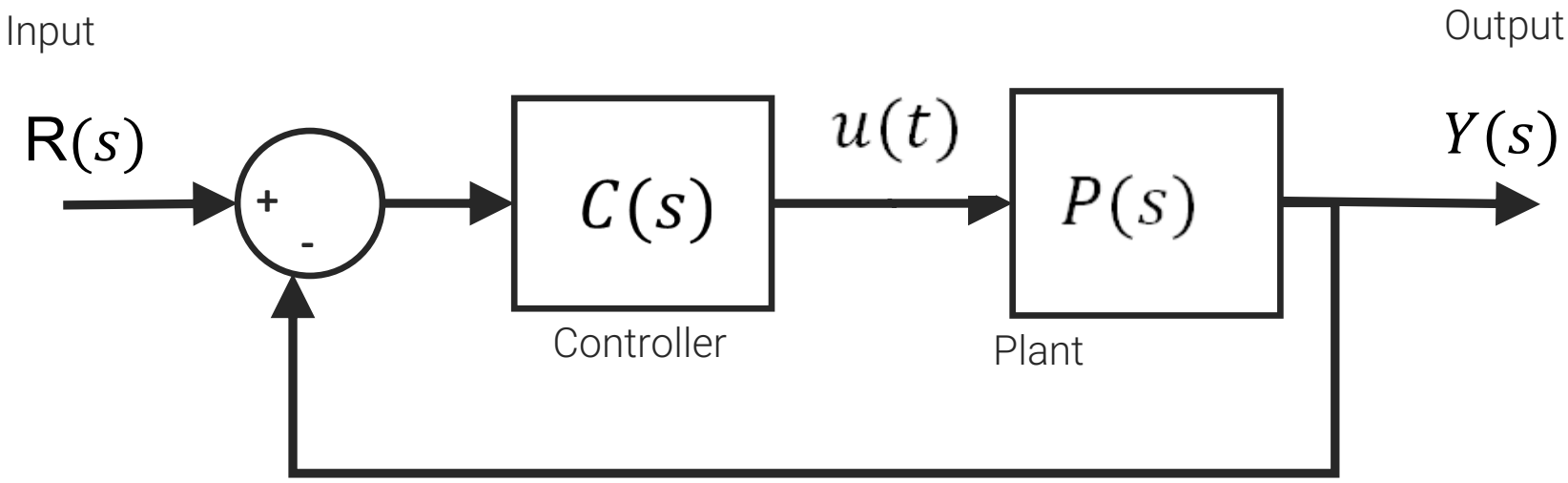


Closed-loop system

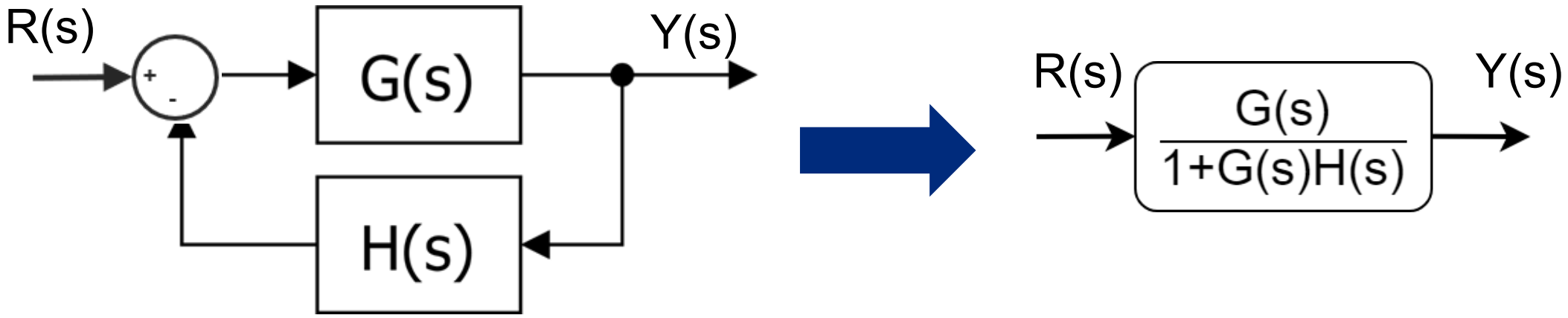


Unity Feedback Closed-Loop System

Closed-loop system with unity feedback



Transfer function: Closed-Loop System



$$\frac{\mathrm{Y}(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

Block Diagrams: Takeaways

Benefits

- Clearly shows how different elements in a system interact with each other
- Can help simplify transfer functions

Drawbacks

- Need to know the system dynamics well
- Can be a tedious process to show how all system components interact with each other

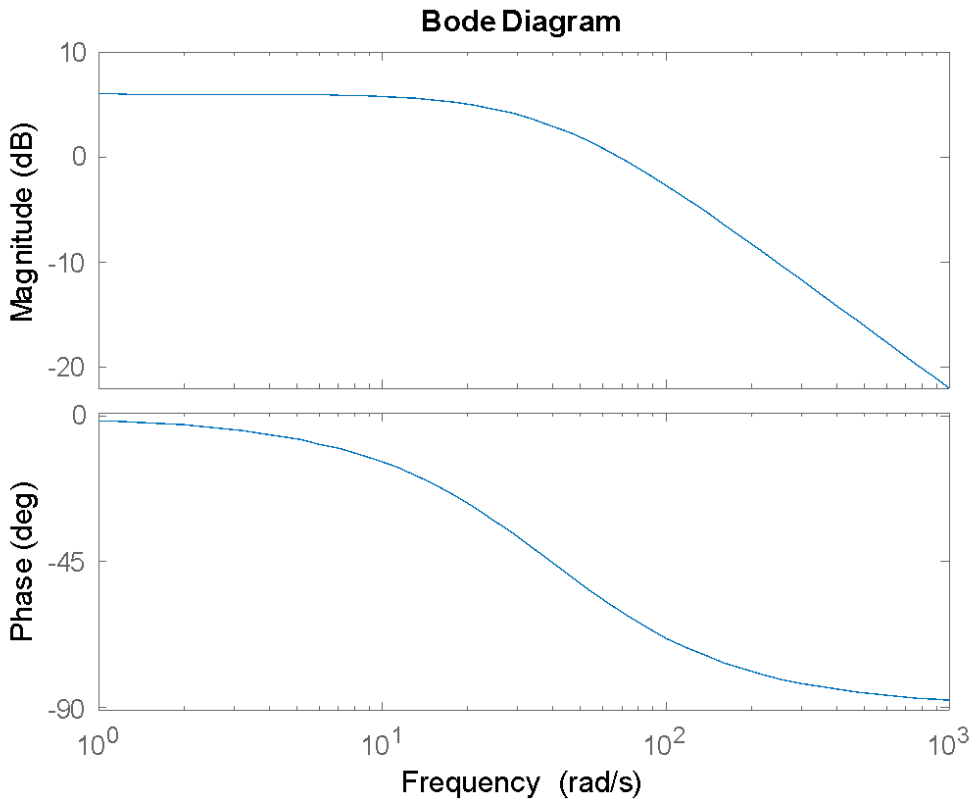
Frequency Response Modelling

Book “Control Systems Engineering”, Chapter 10

What is a *frequency response* model?

- Describe a linear system using sinusoidal inputs and observing the output

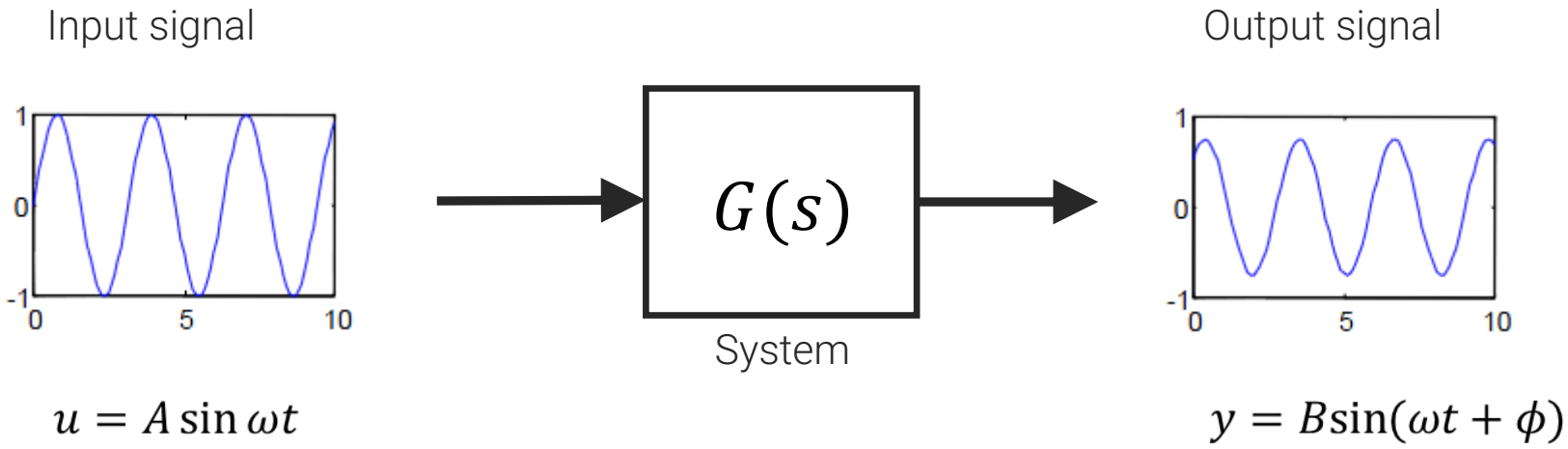
- You end up with two frequency response plots with gain and phase components



Frequency response of a system

Frequency Response Modelling

Relationship between the gain and phase of a sinusoidal input and output response.



Gain of the system:

$$|G(j\omega)| = \frac{|y(\omega)|}{|u(\omega)|}$$

Phase of the system:

$$\phi = \angle G(j\omega)$$

Input-Output Response

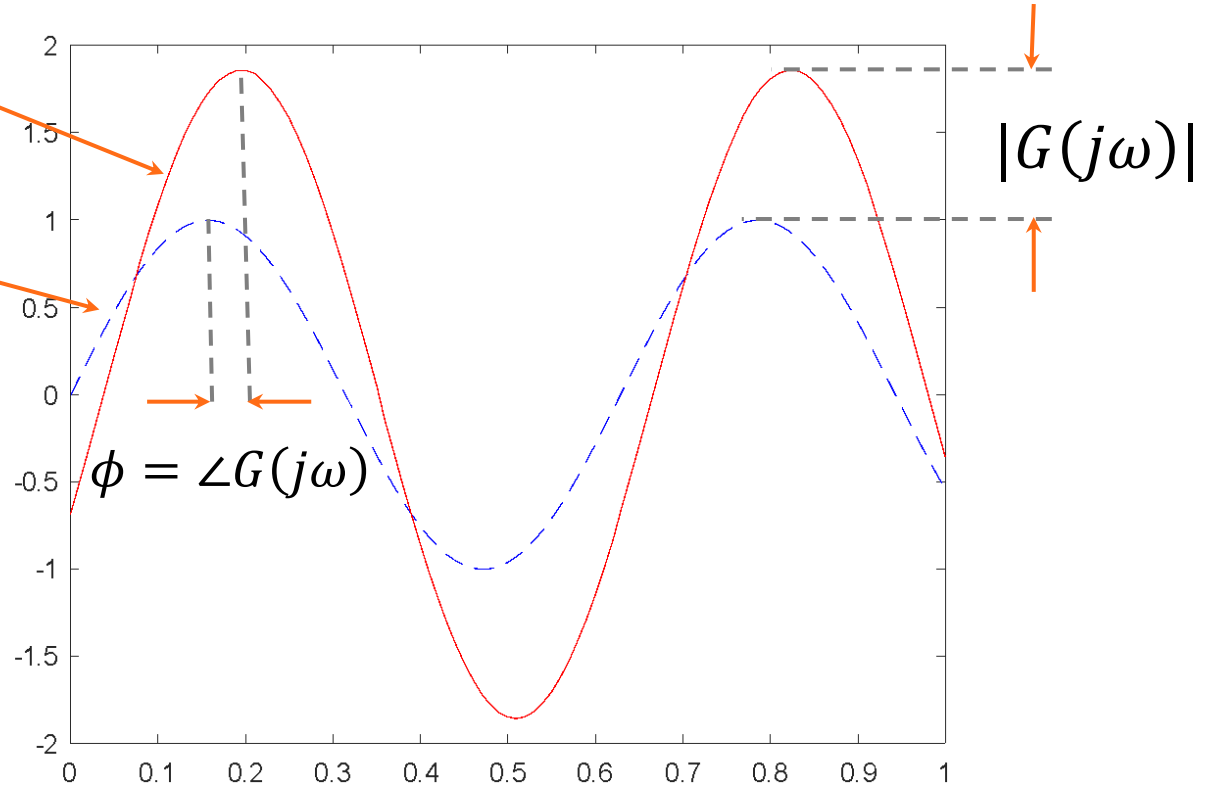
Gain and Phase of output depend on system dynamics of $G(s)$

Output:

$y = A|G(j\omega)| \sin(\omega t + \angle G(j\omega))$

Input:

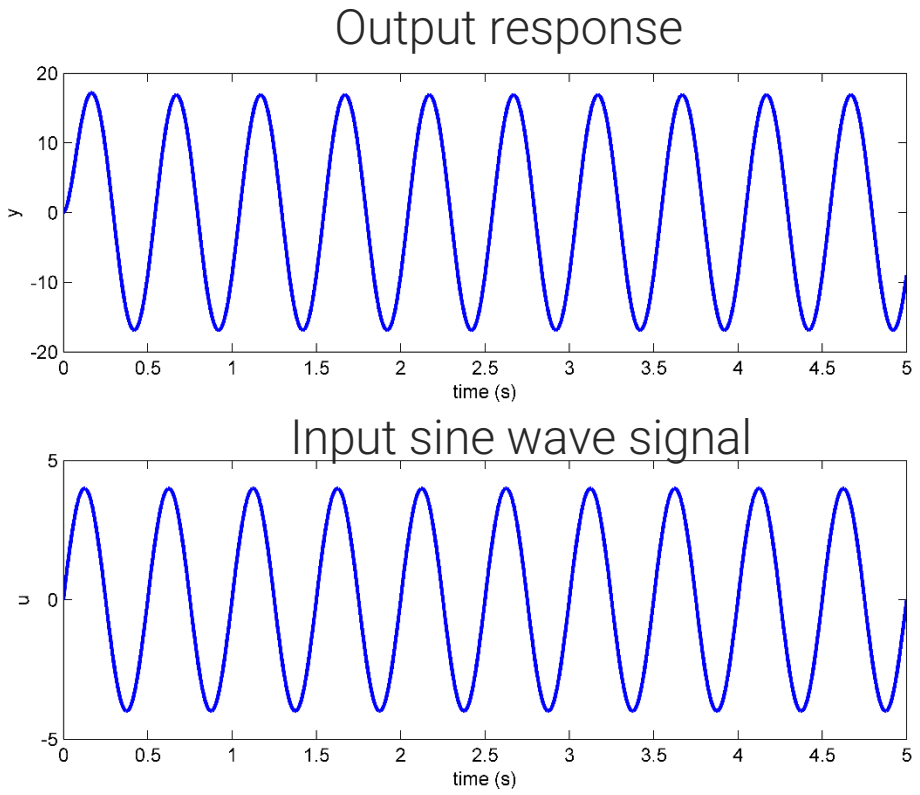
$u = \sin \omega t$



Finding frequency response experimentally

The frequency response of a system can be found *experimentally* by:

1. Applying a sine wave at different frequencies,
2. Examining the resulting output
3. Plot the gain vs. frequency to obtain the gain frequency response plot

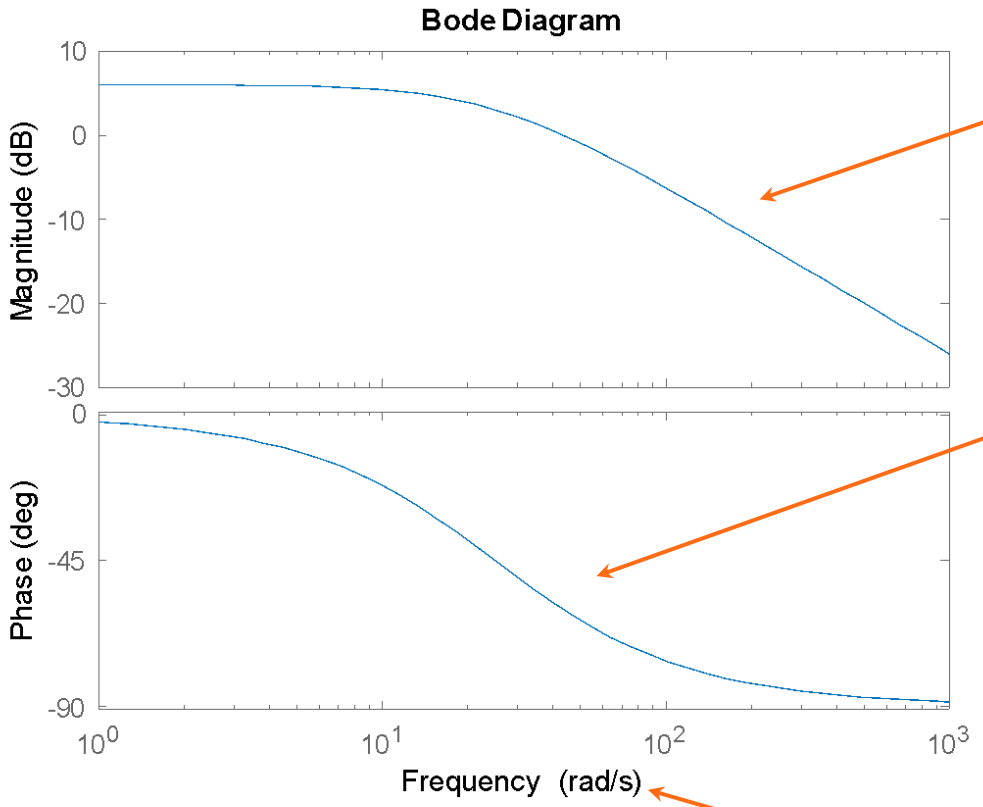


Bode plots

Frequency response plot is known as the *Bode plots of a system.*

Decibel dB
$\mathbf{20\log_{10}|G(j\omega)|}$ **dB** is a logarithmic unit, used to express the ratio between two values of a physical quantity.

Decade ($\log_{10}\omega$): is a unit for measuring frequency ratios on a logarithmic scale, with one decade corresponding to a ratio of 10 between two frequencies.



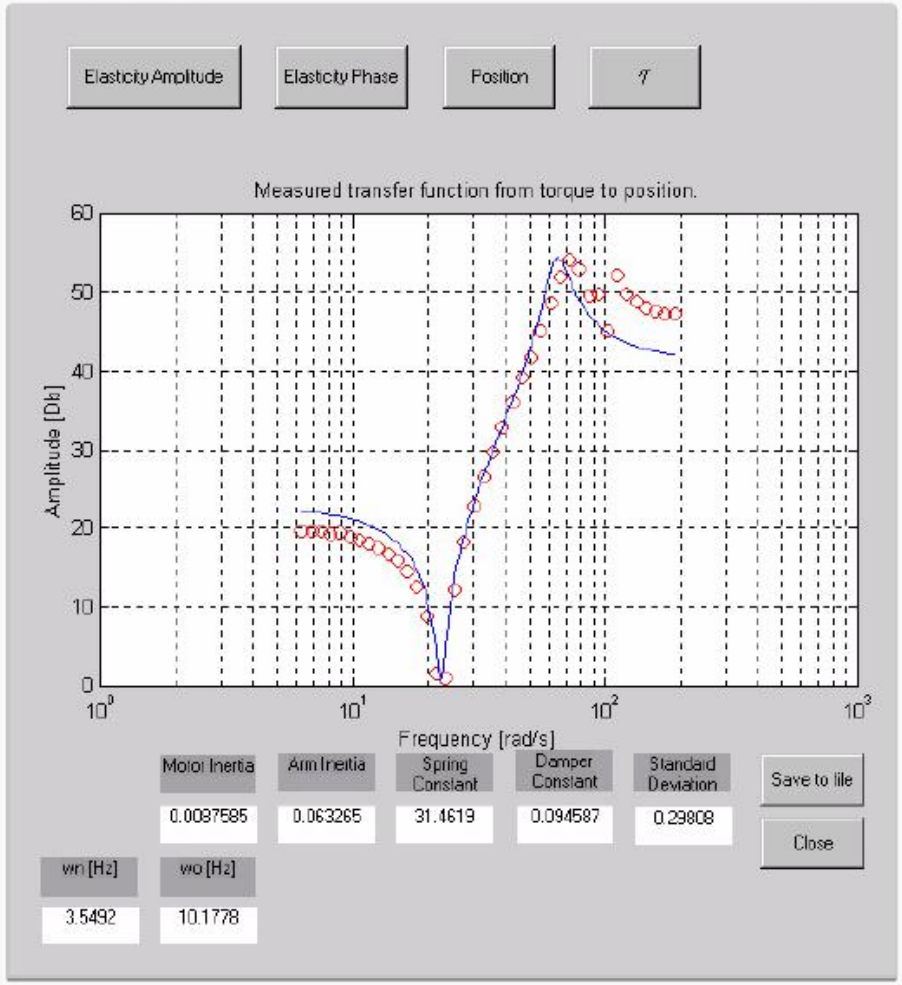
Plot gain of system in dB vs. frequency
$|G(j\omega)|_{dB} = 20\log_{10}|G(j\omega)|$

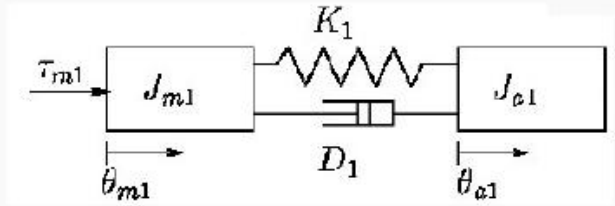
Plot phase φ vs. frequency

$\log_{10}\omega$

Plotting Frequency Response

Example: Magnitude Response for ABB robot IRB 7600





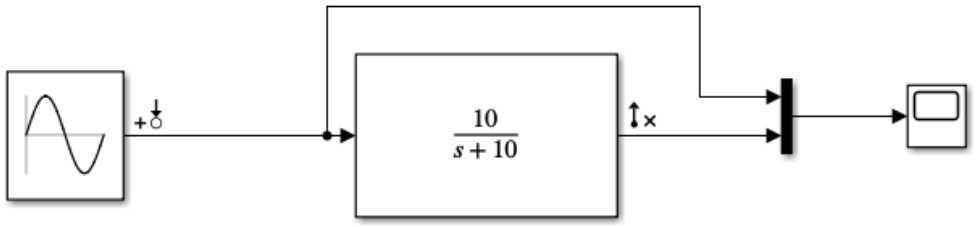


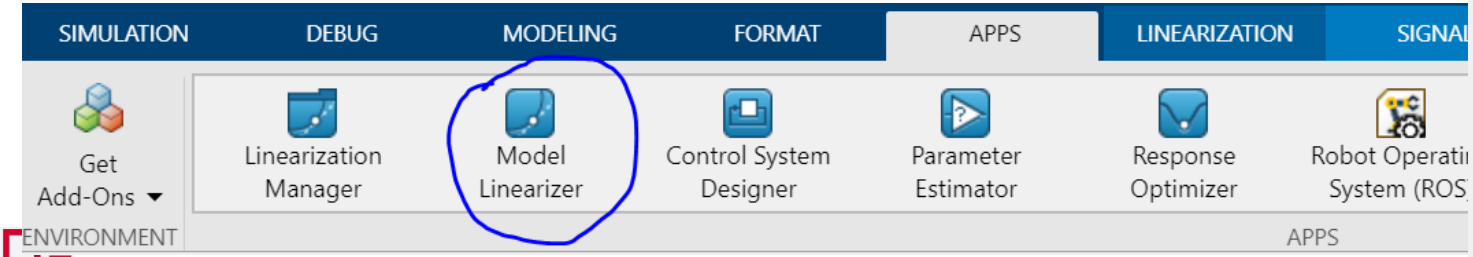
Matlab Bode Plots

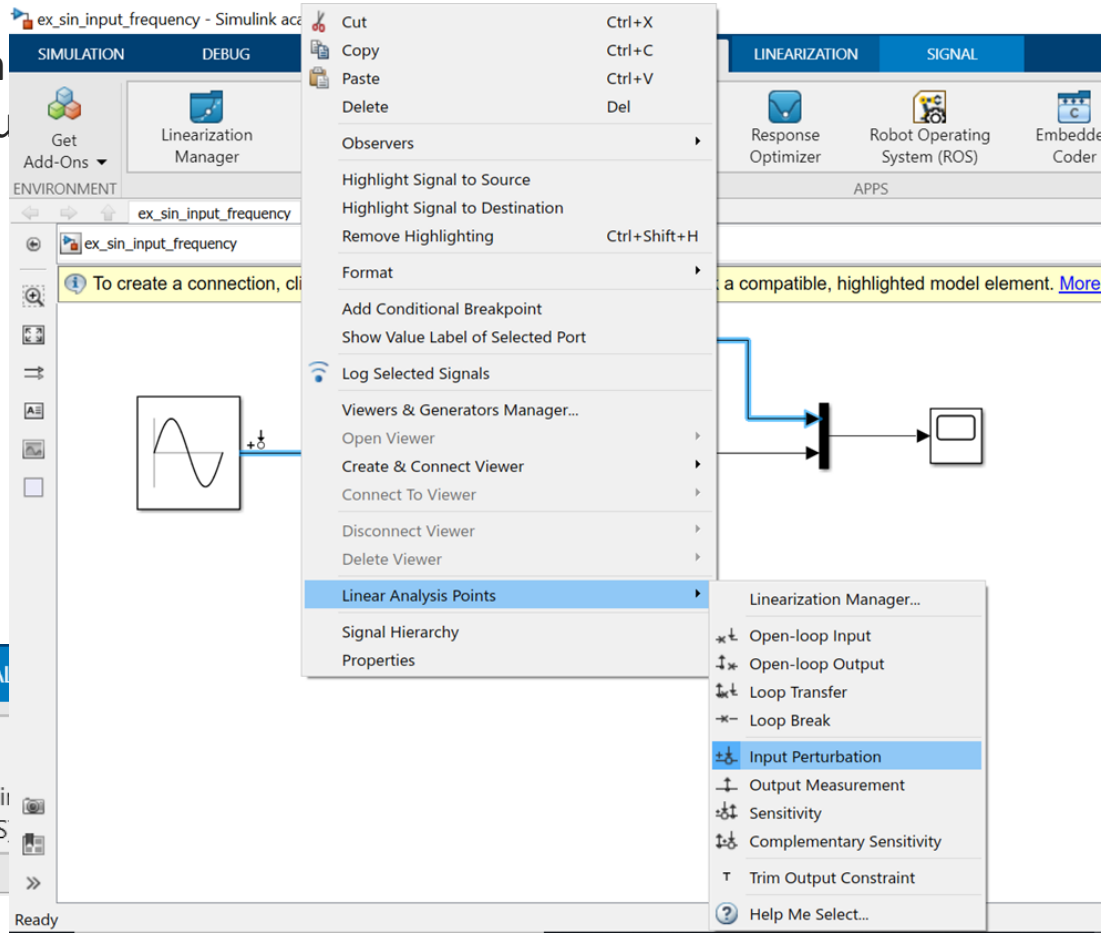
G = tf(10, [1 10]); bode(G)

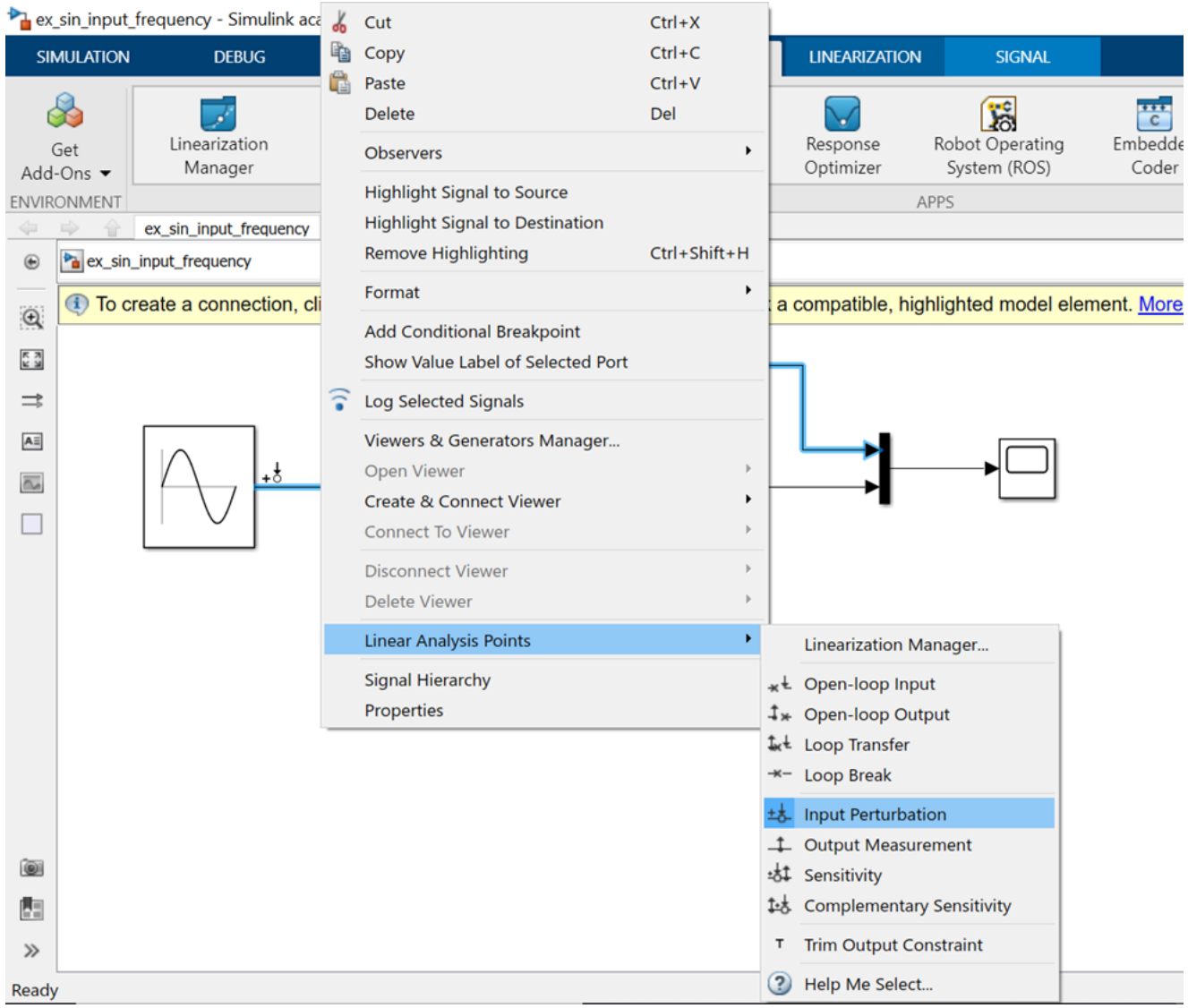
Simulink model:
Right click on the input line, Linear Analysis points/Input Perturbation
Right click on the output line, Linear Analysis points/Open-loop Outpu

APPS/Mode Linearization








ex_sin_input_frequency - Simulink aca
SIMULATION
DEBUG
LINEARIZATION
SIGNAL
Get Add-Ons
Linearization Manager
Response Optimizer
Robot Operating System (ROS)
Embedde Coder
ENVIRONMENT
APPS
ex_sin_input_frequency
ex_sin_input_frequency
To create a connection, cli
a compatible, highlighted model element. More
Cut Ctrl+X
Copy Ctrl+C
Paste Ctrl+V
Delete Del
Observers
Highlight Signal to Source
Highlight Signal to Destination
Remove Highlighting Ctrl+Shift+H
Format
Add Conditional Breakpoint
Show Value Label of Selected Port
Log Selected Signals
Viewers & Generators Manager...
Open Viewer
Create & Connect Viewer
Connect To Viewer
Disconnect Viewer
Delete Viewer
Linear Analysis Points
Signal Hierarchy
Properties
Linearization Manager...
Open-loop Input
Open-loop Output
Loop Transfer
Loop Break
Input Perturbation
Output Measurement
Sensitivity
Complementary Sensitivity
Trim Output Constraint
Help Me Select...
Ready

Frequency Response Modelling: Takeaways

Benefits

- Frequency response can be used to obtain a system model experimentally
- Can be used to model complex systems that may be difficult to model theoretically
- Use frequency-based control design techniques (lead-lag)

Drawbacks

- Can be tedious process
  - Sometimes more efficient to use other modelling methods
- Cannot be used for unstable or dangerous systems, e.g. cart on track would run off track